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ARTICLE

Hexagonal boron nitride inducing anionic trapping in polyethylene oxide based solid polymer electrolyte for lithium dendrite inhibition

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Here we prepare the hexagonal boron nitride (h-BN)-polyethylene oxide composite polymer electrolyte via a convenient casting method, which reflects high mechanical strength. Meanwhile, the electrochemical properties (electrochemical window and lithium ion transference number) are enhanced but the ionic conductivity of h-BN composite electrolyte is decreased after adding h-BN. Density functional theory (DFT) calculations results show that a stronger binding effect is observed in TFSI⁻ and BN, compared to that between Li⁺ and BN. Molecular dynamics (MD) simulations are also utilized to study the mechanism behind the enhanced Li ionic diffusion by h-BN addition. Li⁺ diffusion in PEO/LiTFSI/BN is lower than that in PEO/LiTFSI system, but the diffusion of TFSI⁻ exhibits a more significant decline rate in the presence of BN. It indicates that the presence of BN suppresses anion motion and enhance selectivity in Li⁺ transport. Thus, the PEO/LiTFSI/h-BN composite electrolyte exhibits higher Li ion conductivity but lower anionic diffusivity than PEO/LiTFSI system. Hence the h-BN composite polymer electrolyte in a Li/Li symmetric battery reflects long cycling of 430 h at 0.2 mA cm⁻². The Li metal/LiFePO₄ full battery with PEO/LiTFSI/h-BN composite electrolyte also works efficiently for long-term cycling (140 cycles) than filler-free PEO based electrolyte (39 cycles).

Introduction

With the development of electric vehicle, electronic device and portable electronic devices, the demand of high-energy-density storage systems increase.^{1–6} However, the energy density of lithium-ion batteries is insufficient for some applications. Lithium metal anode with low reduction potential and high theoretical capacity has been a promising candidate to enhance the energy density of battery.^{7–10} The uncontrolled growth of lithium dendrite not only induces low Coulombic efficiency and deteriorates electrochemical stability, but also potentially cause severe safety issues potentially.^{11–16} Moreover, the conventional organic electrolyte is volatile, which may dry out at high temperature, giving rise to short-circuit.^{17,18} The flammable electrolyte also adds risk of fire when leakage happens.

Designing all-solid polymer lithium metal battery is a promising approach to suppress the undesired growth of lithium dendrites. Particularly, solid polymer electrolytes (SPE) reflect high safety with

a stable interface with Li metal anode.^{19–23} Polyethylene oxide (PEO) is a good candidate as the polymer host for SPE attributed to its low cost, light weight and good miscibility with lithium salts.^{24–26} However, PEO based SPE shows limited ionic conductivity, low Li⁺ transference number, narrow electrochemical stability window as well as poor thermal and electrochemical stability, which leads to unsatisfied performance of lithium-polymer battery using PEO based SPE.^{27–30} Lithium ion conductor (Li₇La₃Zr₂O₁₂ and Li₁₀GeP₂S₁₂)^{31,32} and ionic liquid have been mixed in the SPE system to provide extra Li ion transport. However, the high cost of lithium ion conductor and ionic liquid hinders their application in large scale. Blending inactive inorganic fillers (Al₂O₃, TiO₂, SiO₂)^{33–35} into PEO based SPE is regarded as an effective method to enhance the physicochemical properties of SPE owing to the formation of effective ion transport paths.

The hexagonal boron nitride (h-BN) have already been applied in the gel polymer electrolyte (GPE) as fillers to strengthen electrochemical properties and protect of lithium anode due to its high mechanical strength, electrically insulating property and chemical stability.³⁶ Electron-insulated BN provide large specific surface area that will facilitate high ionic conductivity.³⁷ The h-BN in the gel polymer electrolyte shows positive effects on improving the Li⁺ transference number, suppressing growth of Li dendrite and updating mechanical modulus of GPE,³⁸ in which the suppressing ability of dendrite growth is ascribed to the large adsorption energy of h-BN validated by first-principle calculation. In addition, B atoms of BN exhibit Lewis acidic characteristics, which will interact with Lewis bases, i.e. anions in the electrolyte, resulting in the trapping of

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anions which in turn increase the lithium ion transference number (t_{Li^+}) of SPEs.^{39–41} Though h-BN has all the aforementioned advantages, its application as additive to composite polymer electrolyte (CPE) has not been studied explicitly, no reports have been found on designing, characterization and evaluation of liquid free PEO/LiTFSI/h-BN multifunctional CPE with optimal t_{Li^+} and high mechanical strength.

In this work, PEO based composite solid polymer electrolyte (CPE) containing hexagonal boron nitride (h-BN) and lithium salt were prepared, and the effect of h-BN on the mechanical property, physiochemical behavior as well as electrochemical properties of CPE was investigated. To the best of our knowledge, this is the first report on this novel PEO/Li salt/h-BN system. The host material (PEO) were also used as the binder for the cathode in the batteries. The formation and growth of lithium dendrites is expected to be suppressed due to uniformly distributed lithium induced by h-BN. The tensile modulus of CPE and lithium ion transference number are enhanced simultaneously by the addition of h-BN. Density functional theory (DFT) calculations and molecular dynamics (MD) simulations were utilized to explore the mechanism of improved lithium ion transference number and decreased overall ionic conductivity. The mechanism of suppressed anion motion and enhanced cation ion transport are discussed. The simulations and calculation results suggest that the presence of h-BN improve the Li ion conductivity of the composite electrolyte. The superiority of PEO/LiTFSI/h-BN CPE has been testified by all-solid state Li battery using CPE as electrolyte. The designed all-solid-state polymer Li metal/LiFePO₄ with PEO/LiTFSI/h-BN composite electrolyte exhibited a high initial capacity (143.3 mAh g⁻¹) and capacity retention (93 %) after 140 cycles. Meanwhile, the Li/CPE/Li can be stable for 430 h at 0.2 mA cm⁻² due to the stable interface contact.

Experimental Section

Preparation of Composite Polymer Electrolyte (CPE): PEO (molecular weight: 300,000) (Aladdin), Lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) (aladdin) and h-BN (Linker company) were used without further purification. Certain amounts of LiTFSI, h-BN and PEO were loaded in acetonitrile (CH₃CN) and stirred for 5 h to form a homogenized solution as precursor of CPE. CPE membranes were prepared by casting precursor in a mold. The investigated molar ratios of EO/Li salts were 20:1, 32:1 and 44:1, and the content of h-BN in the CPE varied from 0 to 9 wt%.

Materials characterization: Scanning electron microscopy (SEM, Gemini SEM 500) was used to study the morphology. Fourier transform infrared (FT-IR) spectra were recorded on a VERTEX70 (Bruker) spectrometer. Composition of CPEs were investigated by X-ray diffraction (XRD) (Bruker). The samples were scanned in a 2θ range from 10° to 70°. The thermochemical properties were analyzed using a differential scanning calorimeter (DSC, TA Q2000) in the temperature range between -70 °C and 100 °C with a scanning rate of 10 °C min⁻¹ under N₂ atmosphere. The mechanical tests were performed on an electronic tensile machine (CMT6503, MTS).

Electrochemical Measurements: The ionic conductivity of the composite polymer electrolytes was evaluated by electrochemical

impedance technique with electrochemical working station (CHI660E). The SPE/CPE films were sandwiched between two parallel stainless steel discs. The ionic conductivity (σ) of electrolyte membranes is calculated by equation (1), in which R_b represents the resistance of the electrolytes, L the thickness (cm) and S the area (cm²) of the solid electrolytes, respectively. The resistance R_b was measured by electrochemical impedance with at the open-circuit voltage in the frequency range of 1000 kHz to 0.1 Hz with amplitude of 5 mV in the temperature range of 30 °C and 80 °C.

$$\sigma = \frac{L}{R_b S} \quad (2)$$

The electrochemical window of the CPE was tested by linear sweep voltammetry (LSV) technique in a cell that sandwiched the electrolyte film between stainless steel and lithium metal. LSV was performed at 60 °C on a CHI660E electrochemical workstation with scanning rate of 1 mV s⁻¹.

The lithium ion transference number (t_{Li^+}) at 60 °C, t_{Li^+} , was calculated by equation (2):

$$t_{\text{Li}^+} = \frac{I_s \Delta V - I_0 R_0}{I_0 \Delta V - I_s R_s} \quad (3)$$

The initial current I_0 and the steady-state current I_s of a symmetrical Li/CPE/Li cell were measured with a voltage pulse (ΔV) of 10 mV by DC polarization. Electrochemical impedance spectroscopy was conducted before and after the DC polarization measurement to detect the initial and the final interfacial resistances, R_0 and R_s .

Battery Testing: The electrochemical tests were performed under 60 °C via coin cell with lithium metal anode, SPE and LiFePO₄ cathode (working electrode). The working electrode consisted of active material (LiFePO₄), PEO and conductive carbon black in a 7: 1: 2 weight ratio. PEO was used as binder. The LiFePO₄, PEO and carbon black powders were mixed in a mortar and N-methyl-2-pyrrolidone (NMP) was used for the solvent. The slurry of mixture was coated on aluminum foil and deal with vacuum drying process under 100 °C for 12 h. The active material loading was 1.0 - 1.5 mg cm⁻². The batteries were assembled in a glove box flushed with inert air. The charge/discharge tests on Li/CPE/LiFePO₄ batteries were performed using a NEWARE battery tester from 2.8 to 4.0 V.

DFT calculations: The density functional theory (DFT) calculations were used to calculate the binding energy of B₁₂N₁₂ to Li⁺ and TFSI⁻ by Gaussian16 software and Multiwfn 3.6 program. Our DFT calculations were performed using Gaussian 16 software package⁴² at the level B3LYP-D3/6-31+g(d,p), where D3 means that the Grimme's empirical dispersion correction⁴³ was added to the B3LYP functional.

Molecular dynamics (MD) simulations: All-atom classical molecular dynamics (MD) simulations were performed using LAMMPS⁴⁴ simulation package. For all molecular dynamics simulations, the optimized potentials for liquid simulations all-atom (OPLS-AA)⁴⁵ force field was used. In this work, the uncertainty in atomic charges generated by RESP⁴⁶ approaches were investigated. The RESP charge fitted to the electronic structure calculated by Gaussian 16 and Multiwfn 3.6⁴⁷. The atomic charges of the PEO monomer were calculated based on an oligomer consisting of 20 EO monomers. The systems were initially energy-minimized for 10,000 steps using conjugate gradient algorithm to remove abnormally close contacts between molecules. Subsequently, the generated energy-minimized

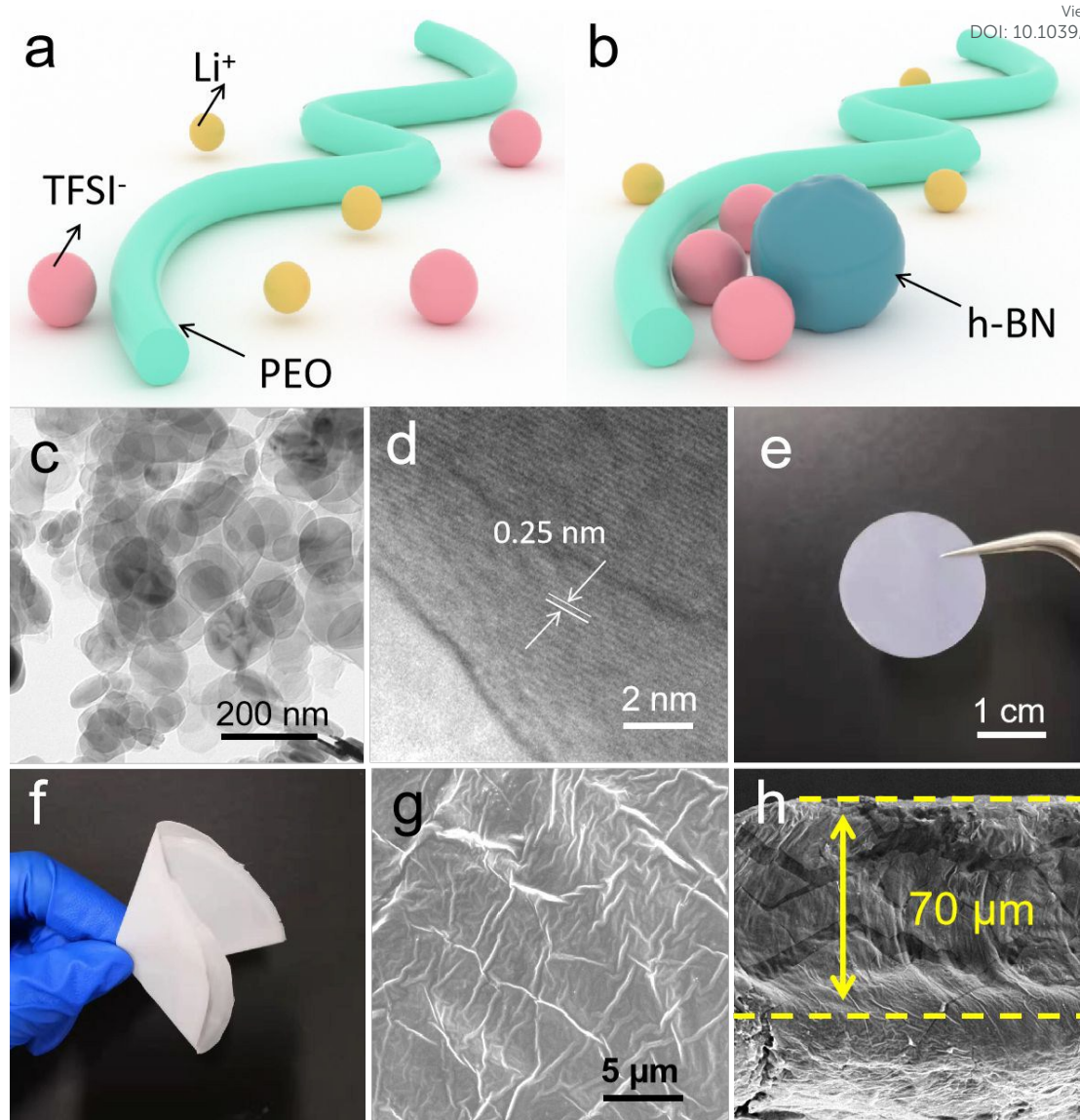


Fig 1. The schematic illustration for the components of PEO/LiTFSI solid polymer electrolyte (a) and PEO/LiTFSI/h-BN composite solid electrolyte membrane (b). (c) The TEM image of h-BN nanosheets. (d) The magnification TEM image of h-BN. (e, f) Photographs of prepared PEO/LiTFSI/h-BN composite polymer electrolyte (CPE) that is a flexible thin film. SEM images: (g) top view and (h) cross-sectional image of PEO/LiTFSI/6% h-BN CPE.

structures were equilibrated at temperature 353 K using the Nosé–Hoover thermostat and Hoover Barostat with relaxation constants of 1 ns, respectively. Mean square displacement (MSD) was calculated for 1 ns using NVT trajectories. The pressure was set at 1.0 atm. The time step is 1 fs. The cutoffs for van der Waals force and the real space of Ewald were 12 Å. The Ewald summation method with a precision of 10^{-6} was used to treat Coulomb interaction in a periodic system.

Results and Discussion

PEO/LiTFSI/h-BN composite polymer electrolyte (CPE) is prepared by the solvent casting method, and schematic illustration for the components of CPE is presented in Fig. 1b. the micro-illustration of PEO/LiTFSI solid polymer electrolyte (SPE) is also shown in Fig.

1a. The h-BN are wright powders and the TEM and SEM images (Fig. 1c and Fig. S1) reflect that the h-BN are well-dispersed ultra-thin nanosheets with diameters ranging from 100–200 nm. In Fig. S2, the elemental mapping of h-BN reflects that the nitrogen and boron elements on h-BN are distributed uniformly. The high-magnification TEM image in Fig. 1d reveals a morphology of clear crystal lattice with an inter-fringe distance of 0.25 nm, indicating a hexagonal structure of BN (JCPDS card: 45-0896). Fig. S3 demonstrates the X-ray diffraction (XRD) pattern of h-BN, which shows the characteristic diffraction peaks corresponding to (002), (100), (004) planes.^{48,49}

After ultrasonic treatment, the h-BN nano-materials are well dispersed in acetonitrile, as can be seen from Fig. S4a. Then, the PEO and LiTFSI are added into h-BN/ acetonitrile mixture under magnetic stirring, a homogeneous white solution is obtained (shown

in Fig. S4b). The as-prepared PEO/LiTFSI/h-BN acetonitrile solution was then casted into a PTFE mould, and kept under room temperature to evaporate the solvent. A free-standing CPE membrane could then be obtained as shown in Fig. 1e. Photograph in Fig. 1f shows a folded CPE membrane, the appearance of the CPE membrane after being folded, no crazing or crack can be observed. SEM image in Fig. 1g shows the morphology of PEO/LiTFSI/6% h-BN CPE. The membrane has rough surface, and the folding on the surface suggest the intrinsic characteristics of polymer. SEM cross-sectional image in Fig. 1h indicates that the thickness of PEO/LiTFSI/6% h-BN CPE is $\sim 70\ \mu\text{m}$.

The X-ray diffraction (XRD) data of pure LiTFSI, PEO, PEO/LiTFSI and PEO/LiTFSI/h-BN CPE are shown in Fig. 2a and Fig. S5. The XRD pattern of LiTFSI depicted in Fig. S5 has sharp

diffraction peaks, which are matched with the JCPDS card (No. 01-082-2341), indicating the crystallinity of the ionic salt. The semi-crystalline nature of pure PEO at room temperature is confirmed by the sharp peaks at 19° and 23° in the XRD pattern shown in Fig. 2a. No remarkable characteristic peaks of LiTFSI are observed in the XRD spectra of either PEO/LiTFSI or PEO/LiTFSI/h-BN CPE indicating the absence of LiTFSI crystal in the membrane which is ascribed to the dissociation of LiTFSI into Li^+ and TFSI^- ions in the CPE system. In spectrum of PEO/LiTFSI/h-BN, no obvious characteristics peaks of h-BN are seen due to its high degree of dispersion in CPE. Compared to those of pure PEO, the relative intensity of the PEO peaks in PEO/LiTFSI/h-BN is found to be decreased with the addition of h-BN, suggesting that h-BN could reduce the crystallization of PEO.

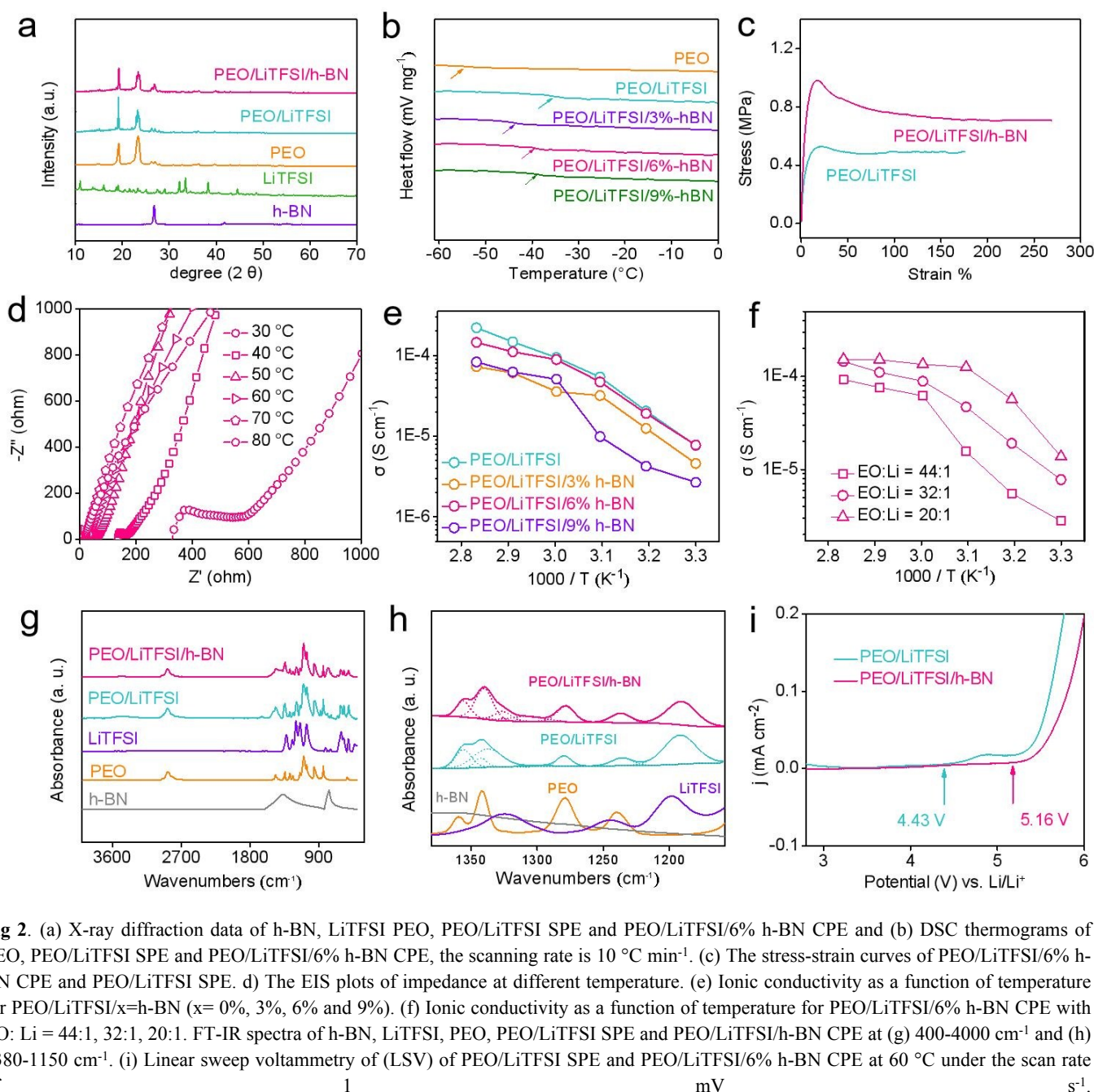


Fig 2. (a) X-ray diffraction data of h-BN, LiTFSI PEO, PEO/LiTFSI SPE and PEO/LiTFSI/6% h-BN CPE and (b) DSC thermograms of PEO, PEO/LiTFSI SPE and PEO/LiTFSI/6% h-BN CPE, the scanning rate is $10\ ^\circ\text{C min}^{-1}$. (c) The stress-strain curves of PEO/LiTFSI/6% h-BN CPE and PEO/LiTFSI SPE. (d) The EIS plots of impedance at different temperature. (e) Ionic conductivity as a function of temperature for PEO/LiTFSI/x=h-BN (x= 0%, 3%, 6% and 9%). (f) Ionic conductivity as a function of temperature for PEO/LiTFSI/6% h-BN CPE with EO: Li = 44:1, 32:1, 20:1. FT-IR spectra of h-BN, LiTFSI, PEO, PEO/LiTFSI SPE and PEO/LiTFSI/h-BN CPE at (g) 400-4000 cm^{-1} and (h) 1380-1150 cm^{-1} . (i) Linear sweep voltammetry of (LSV) of PEO/LiTFSI SPE and PEO/LiTFSI/6% h-BN CPE at $60\ ^\circ\text{C}$ under the scan rate of $1\ \text{mV s}^{-1}$.

Differential scanning calorimetry (DSC) thermograms are employed to validate the thermochemical behavior of PEO matrix by additives. DSC curves of pristine PEO, PEO/LiTFSI SPE and PEO/LiTFSI/h-BN CPE are shown in Fig. 2b. The glass transition temperature (T_g) of pristine PEO, PEO/LiTFSI SPE and PEO/LiTFSI/h-BN CPE are indicated by arrows for each individual curve. T_g of PEO is located at $-56\text{ }^\circ\text{C}$, which shifts to $-36.1\text{ }^\circ\text{C}$ for PEO/LiTFSI SPE with the presence of LiTFSI, indicating the enhanced stiffness of PEO induced by LiTFSI. Compared to T_g of filler-free PEO/LiTFSI SPE ($-36.1\text{ }^\circ\text{C}$), T_g of PEO/LiTFSI/h-BN CPE containing 6wt% h-BN is $-43.9\text{ }^\circ\text{C}$. When the h-BN contents are 3 wt% and 9 wt%, T_g of CPE are $-45.1\text{ }^\circ\text{C}$ and $-38.0\text{ }^\circ\text{C}$. This shift in T_g shows that the chain flexibility in PEO/LiTFSI mixture is promoted by h-BN, indicating the easy movement of ion in the electrolyte system. A single melting transition (T_m) corresponding to PEO in the PEO/LiTFSI/6% h-BN CPE is observed, which indicates that the homogeneous blending of h-BN and PEO has been accomplished without phase separation. The corresponding DSC curves are reflected in Fig. S7a. Thermal stabilities of Celgard 2300, PEO/LiTFSI, and PEO/LiTFSI/h-BN CPE at $25\text{ }^\circ\text{C}$, $60\text{ }^\circ\text{C}$ and $120\text{ }^\circ\text{C}$ are displayed in Fig. S7b. With the temperature increases to $60\text{ }^\circ\text{C}$, the morphology of Celgard 2300 and PEO/LiTFSI film shows sign of thermal shrinkage. In contrast, the morphology of PEO/LiTFSI/6% h-BN CPE is retained well. When the temperature is $120\text{ }^\circ\text{C}$, PEO/LiTFSI/6% h-BN CPE still maintains the pristine state. However, Celgard 2300 rapidly shrinks due to melting transition of polyolefin (PP/PE/PP), and also the melting behavior is observed in PEO/LiTFSI SPE owing to the intrinsic thermal stability, leading to short circuit and explosion of battery at high temperature.

The mechanical strength of polymer electrolyte impacts the electrochemical stability of Li metal full batteries. Owing to its high Young's modulus,³⁶ the stress-strain curves (Fig. 2c) of CPE show that the h-BN filled CPE film exhibits higher tensile strength (0.98 MPa) than that of SPE without h-BN (0.53 MPa). The stress-strain curve in Fig. S8 of PEO/LiTFSI/6% h-BN CPE with molar ratios of EO/Li salts is 20:1, in contrast, has no clear yield point before it breaks, indicating the excessive lithium salts and h-BN lead to degradation of mechanical property.

To confirm the effect of h-BN content on the ionic conductivity of solid polymer electrolyte, electrochemical impedance spectroscopy (EIS) are used to test the ionic conductivity (σ). EIS plots of PEO/LiTFSI/6% h-BN CPE in a frequency range of 10^{-1} - 10^6 Hz and a temperature range from 30 to $80\text{ }^\circ\text{C}$ are reflected in Fig. 2d and Fig. S9. It can be seen from Fig. 2e that for all CPEs, the ionic conductivity exhibits linear dependence to the reciprocal of temperature and can be well fitted to an Arrhenius equation. Moreover, in Fig. 2f, the $\sigma \sim 1000/T$ curves of PEO/LiTFSI/6% h-BN CPE with different EO:Li⁺ (20:1, 32:1, 44:1) from $30\text{ }^\circ\text{C}$ to $80\text{ }^\circ\text{C}$ are displayed. It is clear that increasing amount of ionic salt in CPE system (EO:Li⁺ increases from 44:1 to 20:1) leads to enhanced conductivity of CPE. All values of ionic conductivity are summarized in Table S1 and Table S2. The ionic conductivity of PEO/LiTFSI/6% h-BN CPE is $7.7 \times 10^{-6}\text{ S cm}^{-1}$ and $1.45 \times 10^{-4}\text{ S cm}^{-1}$ at $30\text{ }^\circ\text{C}$ and $80\text{ }^\circ\text{C}$, respectively, which is lower than that of the PEO/LiTFSI SPE ($7.7 \times 10^{-6}\text{ S cm}^{-1}$ and $2.20 \times 10^{-4}\text{ S cm}^{-1}$), suggesting reduced ionic conductivity by mixing with h-BN. On one

hand, this is as a result of the insulated nature of h-BN. On the other hand, filler was required to mechanically block the growth of Li dendrites, while sacrificing the ionic conductivity.³⁸

Fourier-transform infrared spectra (FTIR) in Fig. 2g reflect the complete spectra of h-BN, LiTFSI, PEO, PEO/LiTFSI SPE and PEO/LiTFSI/h-BN CPE in the wavenumber range of 4000 - 400 cm^{-1} . Fig. 2h exhibits the detailed spectra of PEO, PEO-LiTFSI SPE and PEO/LiTFSI/h-BN CPE ranging from 1380 to 1150 cm^{-1} .⁵⁰ Three characteristic peaks of PEO at 1342 and 1279 cm^{-1} (assigned to $-\text{CH}_2-$ wagging absorptions in trans planar structure and $-\text{CH}_2-$ twist in helical structure respectively) can also be observed in spectra of PEO/LiTFSI SPE and PEO/LiTFSI/h-BN CPE. In Fig. 2h, the peak at 1198 cm^{-1} in spectrum of virgin LiTFSI is assigned to the asymmetric stretching of $-\text{CF}_3$. This peak negatively shifts to 1192 cm^{-1} in SPE and CPE, which can be attributed to the dissociation of LiTFSI in PEO. The peak associated to $-\text{SO}_2-$ asymmetric stretching shifts from 1336 cm^{-1} for PEO/LiTFSI SPE to 1326 cm^{-1} for PEO/LiTFSI/h-BN due to the interaction between the $-\text{SO}_2-$ group in the TFSI⁻ anion and the h-BN.^{50,51} The shift of specific peaks testifies the interaction between TFSI⁻ anion and h-BN, which results in the dissociation of LiTFSI and in turn leads to the enhanced release of Li⁺. The additive h-BN can not only react with LiTFSI to release more Li⁺, but also reduce the crystallization degree of PEO and enhance the segmental motion of PEO and further promote the migration of lithium ions.

As additives, h-BN significantly enhance the mobility of polymer chains and increase the Li ion conductivity and the mechanical properties of electrolyte membranes. To investigate the electrochemical window of the solid polymer electrolyte, Linear Sweep Voltammetry (LSV) measurements on a Li/CPE/stainless steel batteries at $60\text{ }^\circ\text{C}$ were performed. As shown in Fig. 2i, the oxidative potential of PEO/LiTFSI SPE is 4.43 V vs. Li/Li⁺ at the scan rate of 1 mV s^{-1} . The electrochemical window of PEO/LiTFSI/6% h-BN CPE is 5.16 V , suggesting that the h-BN helps broadening the electrochemical window of PEO solid electrolyte.

A solid electrolyte with high modulus is considered as one of the most promising approaches to improve lithium ion transference number (t_{Li^+}). t_{Li^+} is the major factor to solid polymer electrolyte. AC impedance and chronoamperometry are performed to measure t_{Li^+} of PEO/LiTFSI and PEO/LiTFSI/h-BN in assembled the Li/CPE/Li battery. Current-time curves of the DC-polarization and AC impedance responses (initial and steady-state current conditions) of PEO/LiTFSI/6% h-BN CPE and PEO/LiTFSI SPE are reflected in Fig. 3a and Fig. S10, respectively. According to Bruce-Vincent Equation,⁴⁰ the t_{Li^+} of PEO/LiTFSI/3%-h-BN, PEO/LiTFSI/6%-h-BN and PEO/LiTFSI/9%-h-BN CPE are calculated to be 0.56, 0.33, and 0.29, which are higher than that of PEO/LiTFSI SPE (0.25).

Owing to the predominant transport of negative charges, the Li⁺ transference number of solid polymer electrolyte is low, leading to the growth of dendritic Li and poor utilization of active materials. The strategies of anion traps are used to enhance Li-ion transport. h-BN is considered as one of the efficient anion traps for Li salts. In our case, the assumption is that BN traps the TFSI⁻ anions through interacting with SO_2 groups in TFSI⁻ and thus suppresses the mobility of negative TFSI⁻, synergistically the mobility of Li⁺ is enhanced by the dissociation of LiTFSI. Therefore, BN composite

achieves low anionic diffusivity but sufficient Li-ion conductivity.

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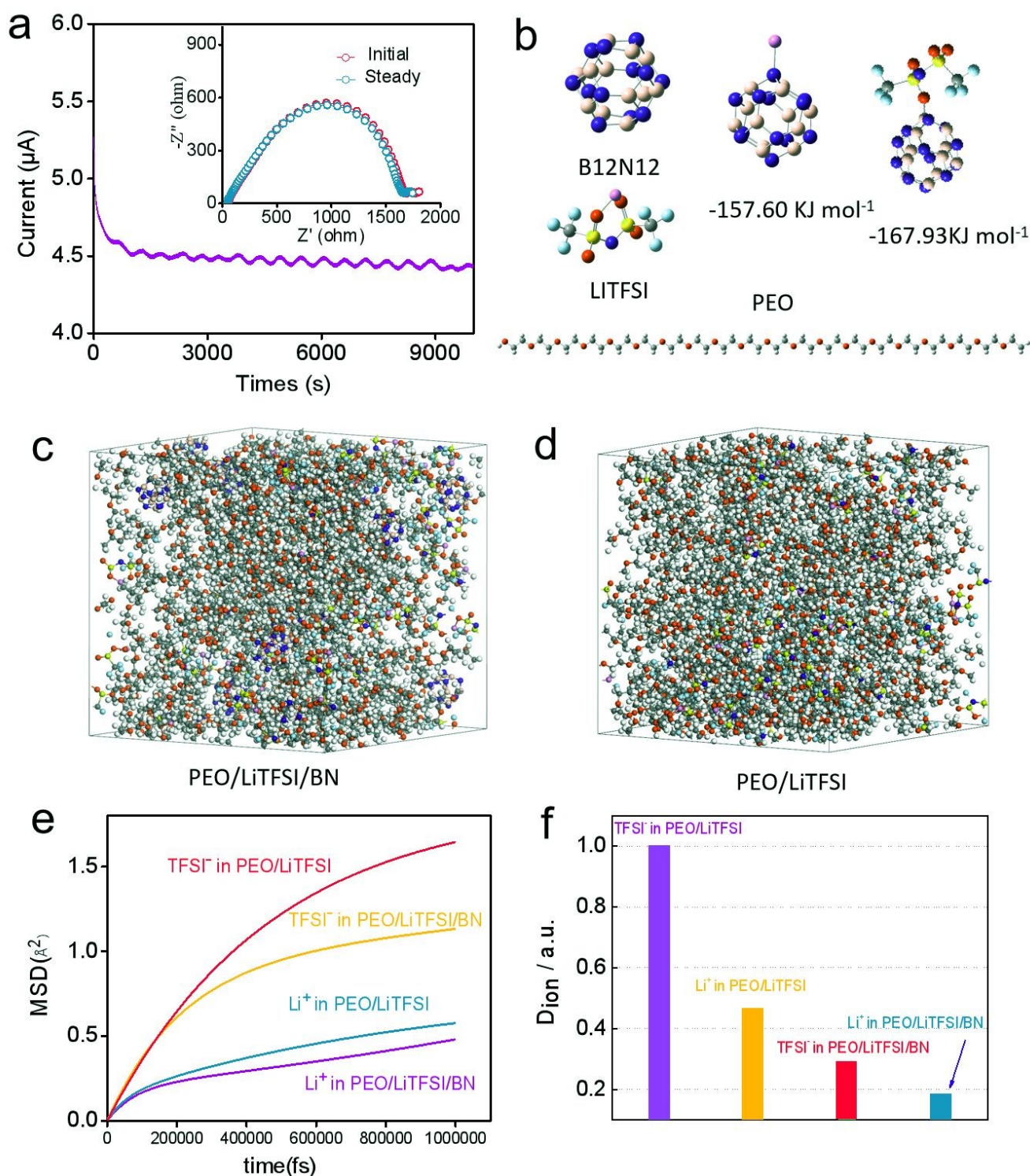


Fig 3. (a) Chronoamperometry curve for PEO/LiTFSI/6% h-BN CPE at 60 °C with polarization voltage of 10 mV and EIS plots before and after the polarization inset. (b) The optimized configuration of LiTFSI/BN complex. Simulation model of Li^+ transport in (c) PEO/LiTFSI and (d) PEO/LiTFSI/BN system. (e) MSD versus diffusion time of Li transport in PEO systems on log scale. MD analysis of lithium-ion diffusion in PEO. (f) The diffusion coefficients of TFSI^- and Li^+ in PEO/LiTFSI and PEO/LiTFSI/BN system.

To validate the above assumption, the density functional theory (DFT) calculations and molecular dynamics (MD) simulations were combined to study the reaction mechanisms in PEO/LiTFSI/BN.^{52–55}

DFT calculations were performed to investigate the interactions between BN and LiTFSI and its effect on enhancing Li^+ transport and suppressing mobility of TFSI^- simultaneously. The model

molecule of BN has a $B_{12}N_{12}$ pattern, which is prone to aggregation. The configuration of Li^+ and BN is shown in Fig. 3b, and the binding energy is $-157.60 \text{ kJ mol}^{-1}$ between Li and BN. Whereas the binding energy between TFSI⁻ and BN is $-166.51 \text{ kJ mol}^{-1}$, lower than that of Li and BN, confirming a stronger binding between TFSI⁻ and BN comparing to Li-BN bond. The calculation results demonstrate that the lithium ion transfer number is promoted owing to the restrained TFSI⁻ transport by BN trapping, which is in good consistence to the experimental results. The molecular dynamics simulations are employed to investigate the mechanism of decreased anionic diffusivity and enhanced lithium ion transference of BN gifted PEO electrolyte.^{56–59} The mobility of Li^+ in PEO/LiTFSI and PEO/LiTFSI/BN system are compared, and the simulation snapshots are reflected in c and d. Mean squared displacements (MSDs) and diffusion coefficients of Li^+ and TFSI⁻ ions in PEO/LiTFSI and

PEO/LiTFSI/BN mixtures are calculated and revealed in Fig. 3e and f. The Li^+ diffusion in PEO/LiTFSI/BN is observed to be lower than that in PEO/LiTFSI system as shown in Fig. 3f. Compared to the change of Li diffusion in systems with and without BN, the diffusion of TFSI⁻ exhibits a more significant decline rate in the presence of BN, indicating the prominent effect of BN on the diffusion of TFSI⁻. The presence of BN suppresses the anion motion and enhance selectivity in Li^+ transport as well as decreased cation mobility as shown in Fig. 1b.^{60–63} These simulation results illustrate that the TFSI conductivity is more decreased in the PEO/LiTFSI/BN system than Li ionic conductivity in the same system by adding BN. This is the reason why the overall ionic conductivity of BN composite solid electrolyte decreased and the Li ionic conductivity are enhanced comparing with PEO/LiTFSI electrolyte.

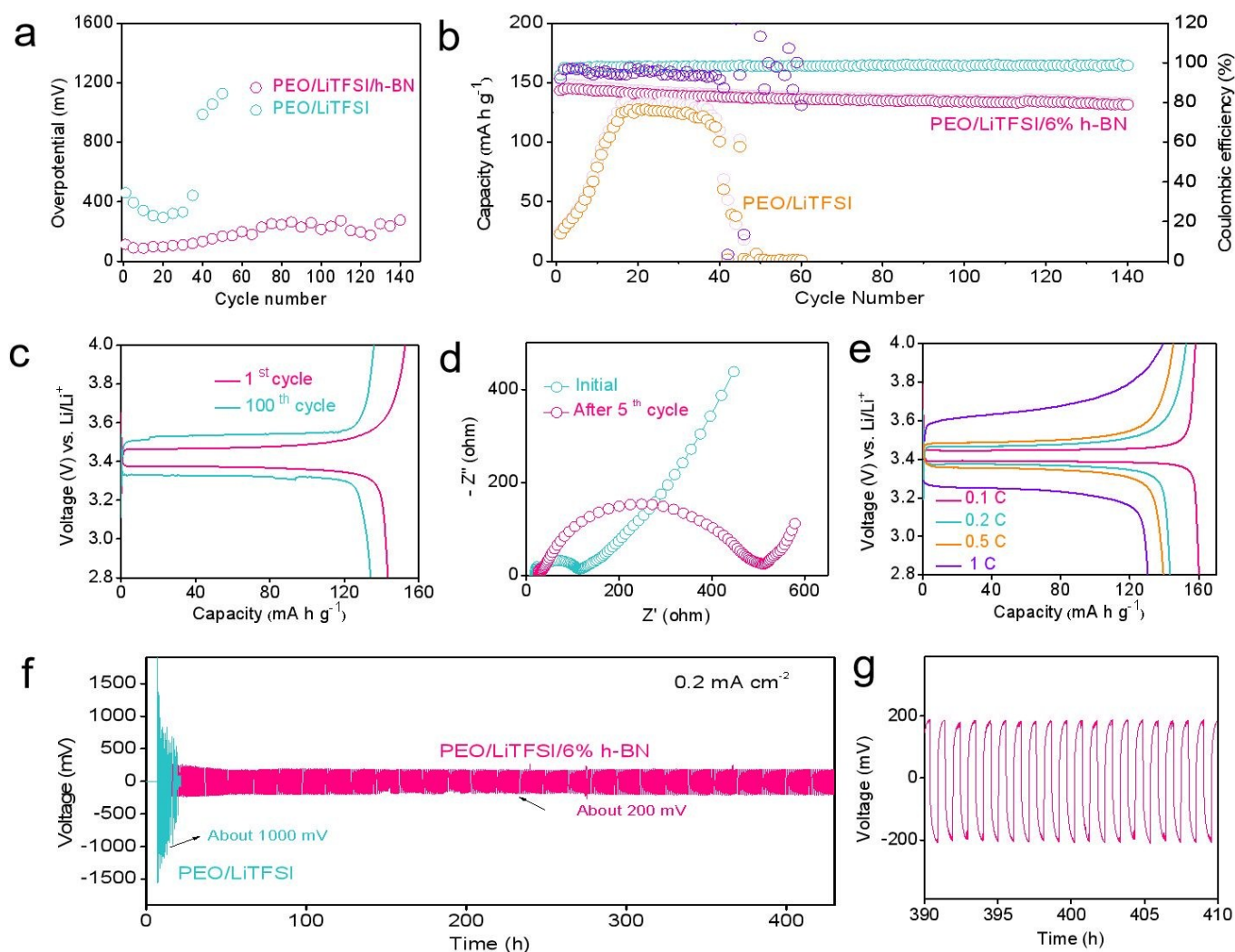


Fig 4. (a) Overpotential of the battery with PEO/LiTFSI/6% h-BN CPE and PEO/LiTFSI SPE after 140 cycles. (b) Long term cycle performance of Li metal battery with PEO/LiTFSI/6% h-BN CPE and PEO/LiTFSI SPE at 60°C . (c) Charge/discharge voltage curves for 1st and 100th cycles of Li/PEO/LiTFSI/6% h-BN/LiFePO₄ at a current rate of 0.2 C at 60°C . (d) Impedance spectra of the Li/PEO/LiTFSI/6% h-BN/LiFePO₄ battery before and after 5 cycles. (e) The charge and discharge curves of battery with PEO/LiTFSI/6% h-BN CPE at 0.1C, 0.2 C, 0.5 C, and 1C. (f) Galvanostatic cycling of Li symmetric battery using PEO/LiTFSI/% h-BN CPE and PEO/LiTFSI SPE at the current density of 0.2 mA cm^{-2} . (g) The tail region of Galvanostatic cycling of Li symmetric battery using PEO/LiTFSI/h-BN CPE for selected cycles.

Electrochemical cycle performance of all-solid-state Li metal/LiFePO₄ full battery using PEO/LiTFSI/6% h-BN CPE or PEO/LiTFSI SPE as electrolyte are tested at 0.2 C and 60 °C. In Fig. 4b, the battery using PEO/LiTFSI/6% h-BN delivers higher initial Coulombic efficiency (94 %) than that of PEO/LiTFSI (92 %) because of the higher mobility of Li⁺ in PEO/LiTFSI/6% h-BN. The high capacity retention (93 %) for the battery with PEO/LiTFSI/6% h-BN is reflected after 140 cycles with the capacity remaining at 134 mAh g⁻¹. In comparison, the battery using PEO/LiTFSI SPE shows a lower discharge capacity of 22.7 mAh g⁻¹, due to the high interface resistance between Li metal and electrolyte induced by low Li ion transfer of electrolyte. Overcharging occurs after 39 cycles and the Coulombic efficiency of PEO/LiTFSI SPE decline rapidly due to the internal short circuit. Furthermore, the overpotential values of the Li/PEO/LiTFSI/6% h-BN CPE/LiFePO₄ battery are smaller than that of Li/PEO/LiTFSI/LiFePO₄ battery in overpotential-cycle number profiles (Fig. 4a), demonstrating that the degree of electrode polarization is decreased by the composite polymer electrolyte. Table S3 reflects the comparison of long term cycling performance of Li/PEO/LiTFSI/6% h-BN/LiFePO₄ battery with other reported PEO-based electrolyte Li metal battery in recent years.

The charge-discharge curves at 0.2 C of Li/PEO/LiTFSI/6% h-BN/LiFePO₄ battery are presented in Fig. 4c. Typical discharge and charge plateaus are observed, suggesting the good stability for LiFePO₄ during cycling. The initial discharge and charge capacities are 143.3 mAh g⁻¹ and 152.6 mAh g⁻¹ for the 1st and 100th cycle, respectively. In comparison, the initial capacity of the cell using PEO/LiTFSI electrolyte is only 22.7 mAh g⁻¹ due to the high interface resistance and low ionic conductivity. The potential plateaus of the initial cycle are at 3.47 V for the charging process and 3.36 V for the discharging process for the 1st cycle indicating a small polarization with the voltage difference between charge and discharge only being 110 mV. The voltage difference increased slightly after 140 cycles, further suggest that the PEO/LiTFSI/6% h-BN provides moveable Li⁺ ions and a stable interface between electrolyte and electrode.

Interfacial resistance of Li-metal battery using solid polymer electrolyte is demonstrated via Electrochemical Impedance Spectroscopy (EIS) measurement. The impedance spectra of the Li/PEO/LiTFSI/6% h-BN/LiFePO₄ and Li/PEO/LiTFSI/LiFePO₄ batteries before and after 5 cycles are revealed in Fig. 4d and Fig. S11. To confirm the change of interfacial impedance, an equivalent circuit is performed and the corresponding results are exhibited in Fig. S12 and Table S4. The bulk resistance (R_b) of the battery using PEO/LiTFSI/6% h-BN rises from 20.94 (initial) to 30.69 Ω (after 5 cycles). In addition, though the interfacial resistance (R_{ct}) interfacial resistance (R_{ct}) increase slightly (from 83.34 to 433.8 Ω) after 5 cycles, it is still lower than 447.2 and 1651 Ω of the battery using PEO/LiTFSI the resistance results indicate that a stable interface is formed between PEO/LiTFSI/6% h-BN and electrode during the charge/discharge cycling.

The charge-discharge curves of Li/PEO/LiTFSI/6% h-BN/LiFePO₄ at 0.1 C, 0.2 C, 0.5 C and 1 C are displayed in Fig. 4e. The corresponding capacities are 161, 150, 129 and 95 mAh g⁻¹, respectively. It demonstrates that the battery with PEO/LiTFSI/6% h-BN CPE has good rate performance at kinds of current densities.

Cyclic stable performance and interface stability of PEO/LiTFSI/h-

BN composite electrolyte to lithium metal are investigated by Li/SPE/Li symmetrical battery. Galvanostatic cycling measurements were performed at a current density of 200 μA cm⁻² at 60 °C, and the results are reflected in Fig. 4f. The magnified tail region of Fig. 4f is shown in Fig. 4g and Fig. S12, and it is clear that PEO/LiTFSI/6% h-BN CPE exhibits a low voltage profile (200 mV) under 200 μA cm⁻². Meanwhile, a stable voltage profile is observed, no short circuit and detectable failure can be detected after 430 h attributing to the enhanced tensile modulus and lithium ion transference of composite solid electrolyte, which can restrain the formation of lithium dendrites and improve the cycle stability of the battery. On the contrary, the voltage profile of PEO/LiTFSI SPE is close to 1000 mV at 200 μA cm⁻² and the voltage profile decreases sharply, which is caused by internal short-circuiting due to the growth of Li dendrites leading to short-circuiting in voltage profile of PEO/LiTFSI SPE. The high cycling stability of the battery is attributed to lithium ion transference number and high tensile modulus induced by the addition of h-BN, which can inhibit the nucleation of lithium during deposition.

The longer cycle life of h-BN CPE battery is attributed to the h-BN improved electrochemical stability, mechanical strength, lithium ion transport number and interfacial stability, which also assist suppressing Li dendrite formation. As shown in Fig. 5a, the long term cycling performance of Li metal battery with PEO/LiTFSI/6% h-BN CPE is also tested under large current density (0.5 C). The battery using PEO/LiTFSI/6% h-BN CPE exhibits higher capacity of 139.5 mAh g⁻¹ than that of the battery with PEO/LiTFSI SPE (8.5 mAh g⁻¹). The charge-discharge curves at 0.5 C in Fig. 5b show that the battery with composite solid polymer electrolyte reveals small polarization. Schematic illustration in Fig. 5c reflects that the lithium dendrite could be observed on the surface of lithium anode after cycling in the battery with PEO/LiTFSI SPE. On the contrary, the Li metal anode in the battery with PEO/LiTFSI/6% h-BN CPE exhibits smooth surface after 50 cycles under 0.5 C. To explore the inhibition of lithium dendrite growth by h-BN enhanced solid electrolyte, the Li electrodes after cycling are examined by SEM. The surface morphology of lithium metal anode taken from the solid state batteries using PEO/LiTFSI and PEO/LiTFSI/6% h-BN SEM images after 50th cycles at 0.5 C and 60 °C are reflected in Fig. 5d and Fig. 5e. In the SEM image (Fig. 5d), unlike the anode in PEO/LiTFSI (Fig. 5e), which shows a bumpy irregularity surface, the anode in PEO/LiTFSI/6% h-BN battery retains a smooth surface. The cross-sectional SEM image (Fig. S15a) reveals the obvious lithium dendrite on the cross-section of lithium metal anode, which is taken from the battery using PEO/LiTFSI. However, in Fig. S15b, the lithium metal anode in PEO/LiTFSI/6% h-BN battery after cycling exhibits a smooth surface similar to the pristine electrode before cycling (Fig. S14b). The SEM results demonstrate that 2D h-BN dispersing in the polymer electrolyte still can suppress the dendrite growth for lithium metal owing to its excellent chemical stability, superior insulating property and large tensile modulus that can delay the nucleation of the lithium dendrites as shown in Fig. 5c.³⁶

The lithium metal battery using solid electrolyte exhibits significant improvements for safety issues. We further evaluated the practical applicability of PEO/LiTFSI/h-BN CPE using a Li metal anode and a LiFePO₄ cathode. A photograph of the prepared soft packed Li-metal battery is revealed in Fig. S16, open circuit potential of the

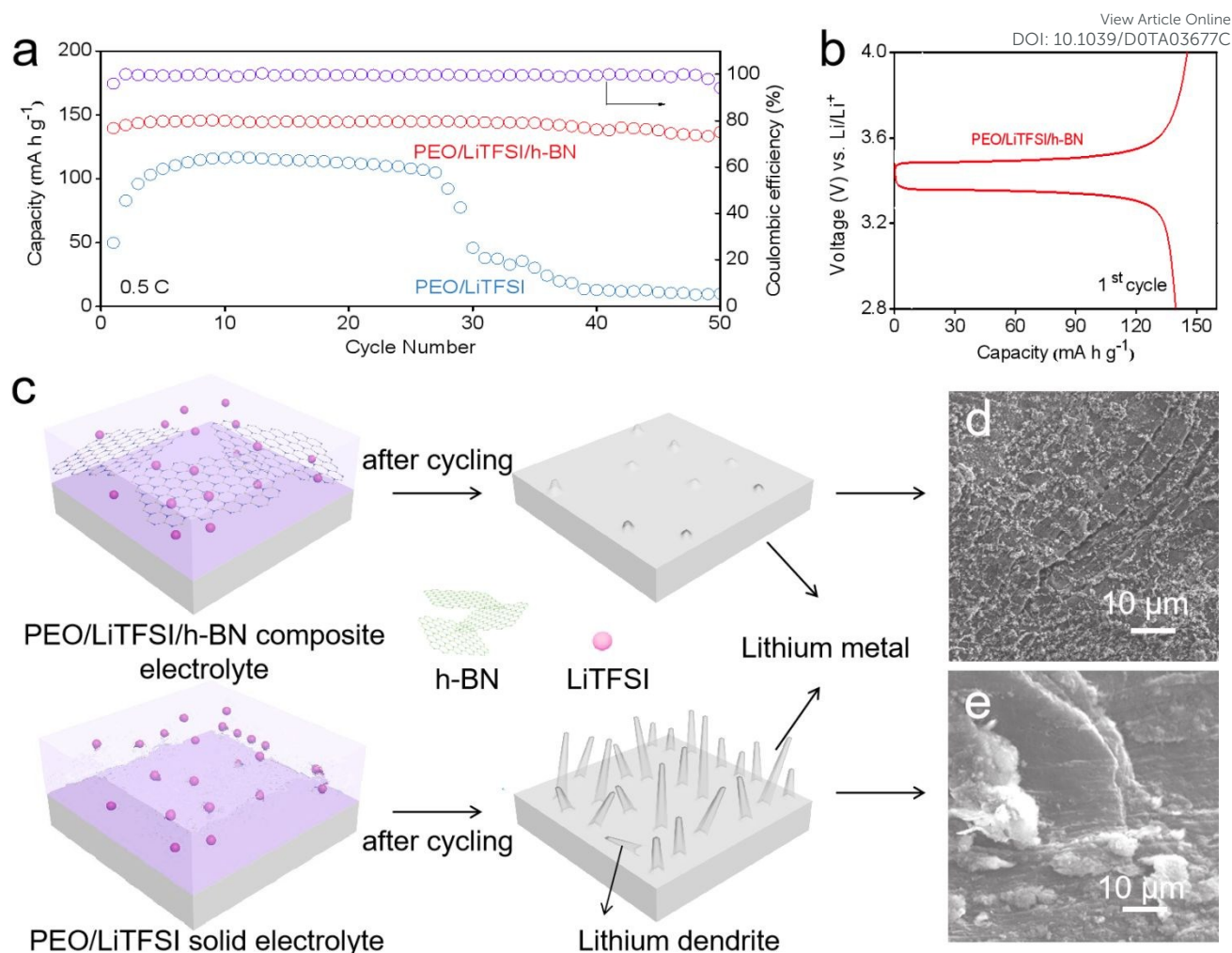


Fig 5. (a) Cycle performance and Coulombic efficiency of Li/PEO/LiTFSI/6% h-BN/LiFePO₄ and Li/PEO/LiTFSI/LiFePO₄ battery at 0.5 C at 60 °C. (b) Charge/discharge voltage curves for 1st cycle of Li/PEO/LiTFSI/6% h-BN/LiFePO₄ and the battery using PEO/LiTFSI SPE at 0.5 C. (c) Schematic showing the lithium dendrite inhibition of PEO/LiTFSI SPE PEO/LiTFSI/6% h-BN CPE after cycling. The SEM images: (d) top view of Li anode after 50 cycles at 0.5 C with PEO/LiTFSI/6% h-BN CPE. (e) top view of Li anode using PEO/LiTFSI SPE.

prepared battery is determined to be 2.23 V. In Fig. S17, charge/discharge of the soft packed solid-state battery have been performed at the voltage ranging from 2.8 to 4.0 V with charging-discharging at 0.1 C current rate under 60 °C. It can be seen that except for the first cycle, the Coulombic efficiency of all the cycles are over 90%. The assembled battery can deliver capacity about 70 - 85 mAh g⁻¹ for 10 cycles. After 10 cycles capacity fading is observed, which is attributed to the absence of assembly pressure in the soft-package battery. Owing to the diffusion of the anion in composite polymer electrolyte is suppressed by adding BN, which mitigates the concentration gradient and polarization, and in turn improves the lithium electrodeposition stability, which can suppress the formation and growth of lithium dendrites in lithium metal batteries.

Conclusion

The h-BN are selected as fillers for preparing PEO-based solid composite electrolyte via a fast and easy method. The as-prepared

PEO/LiTFSI/h-BN CPE exhibits a higher Li ion transference number and higher mechanical tensile property compared to those of solid electrolyte without h-BN. For these reasons, PEO/LiTFSI/h-BN CPE can suppress the growth of lithium dendrites. The DFT results reveal that a stronger binding between TFSI⁻ and BN results in the restrained TFSI⁻ transport and promoted lithium ion transfer simultaneously. MD simulations results suggest that the diffusion of the anion in composite polymer electrolyte is suppressed by adding BN, which mitigates the concentration gradient and polarization, and in turn improves the lithium electrodeposition stability. Benefiting from these enhanced properties, the PEO/LiTFSI/h-BN CPE in a Li/Li symmetric battery shows a long cycling performance for 430 h at 0.2 mA cm⁻². The PEO/LiTFSI/h-BN CPE also works efficiently in Li/LiFePO₄ for long-term cycling performance (140 cycles) with high capacity retention (93 %).

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the

online version at

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Conflicts of interest

There are no conflicts to declare.

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Hexagonal boron nitride inducing anionic trapping in polyethylene oxide based solid polymer electrolyte for lithium dendrite inhibition

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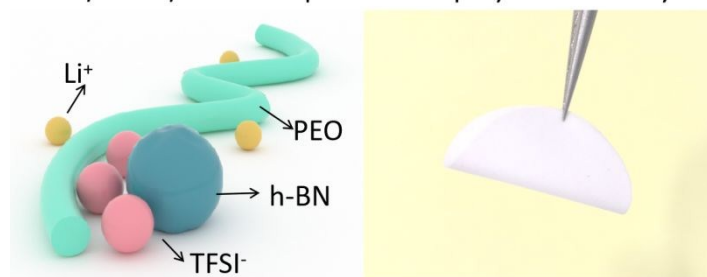
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PEO/LiTFSI/h-BN composite solid polymer electrolyte



Lithium ion conductivity and mechanical strength of PEO based composite solid polymer electrolyte are improved by adding h-BN.